

Here is a comprehensive list of 28 discovery questions, categorized by operational area, designed for a feasibility and needs assessment with a large-scale mining company.

When meeting with a large-scale miner, remember that their answers will likely differ depending on whether you are speaking to the **Lab Manager**, the **IT Director**, or the **Metallurgical/Operations Manager**. These questions are designed to expose the cracks in their current system and validate the need for your disruptive solution.

Category 1: Current State & Pain Points

1. What LIMS or sample tracking software are you currently using across your laboratory operations?
2. On a scale of 1-10, how satisfied are your lab technicians with the day-to-day usability of the current system?
3. What are the top three most frustrating bottlenecks your lab experiences with the current software?
4. How often do manual transcription errors occur, and what is the operational or financial impact when they do?
5. What is your current average Turnaround Time (TAT) from when a sample arrives at the lab to when the assay result is reported to the mill?
6. Are there instances where lab technicians resort to using "shadow IT" (like Excel spreadsheets or paper logs) to bypass the current LIMS? If so, why?

Category 2: Fire Assay & Bullion Workflows

7. Can you walk me through your specific fire assay workflow? At which stages do you currently capture data (e.g., flux mixing, fusion, cupellation, parting, final weighing)?
8. How do you currently calculate flux recipes? Is it a manual process, or is it automated based on the ore matrix?
9. How do you track the lifecycle and consumption of highly sensitive consumables, such as lead oxide, crucibles, and acids?
10. What is your current process for tracking the chain of custody (CoC) for high-grade samples, gold buttons, and final bullion bars?
11. How do you currently generate your Certificates of Analysis (CoA) for bullion bars, and how long does that process take?

12. How do you ensure your bullion bar tracking and documentation meet LBMA (London Bullion Market Association) or ISO 17025 compliance requirements?

Category 3: Hardware & Instrument Integration

13. What specific models of analytical scales, XRF analyzers, and AAS/ICP instruments are you currently running in your labs?
14. Are your instruments currently integrated directly with your LIMS, or are operators manually typing results into the system?
15. If you purchase a new piece of analytical equipment, how difficult and time-consuming is it to integrate it into your current LIMS?
16. What barcode/QR code scanners and label printers are you currently using, and do they hold up well in your lab environment (heat, acid, dust)?

Category 4: Quality Control (QA/QC) & Compliance

17. How are you currently managing QA/QC? (e.g., Do you use Shewhart control charts for blanks, standards, and duplicates?)
18. What happens when a QA/QC sample fails? How is the rejection, re-assay, and exception reporting handled?
19. How easily can your current system produce audit trails for external regulators or internal metallurgical accounting audits (e.g., AMIRA P754)?
20. How do you currently track and flag unusual weight variances in gold buttons or bullion bars that might indicate theft or process loss?

Category 5: IT Infrastructure & Connectivity

21. How reliable is the internet connectivity at your remote mine sites? Do you experience frequent outages?
22. If the internet goes down at a remote site, can your current LIMS continue to operate? What happens to the data captured during the outage?
23. What is your corporate IT policy regarding cloud-hosted SaaS applications versus on-premise enterprise deployments?

24. What hardware do lab technicians use on the bench (e.g., desktop PCs, ruggedized tablets, mounted touchscreens)? Are they able to use the system while wearing PPE (gloves, safety glasses)?
25. How does assay data flow from the lab to your corporate ERP (e.g., SAP, Oracle) or geological modeling software (e.g., Datamine, Leapfrog)?

Category 6: Commercial & Vendor Expectations

26. What is your typical Total Cost of Ownership (TCO) for your current LIMS, considering licensing, maintenance, and required external consultants?
27. What is your typical implementation timeline for new enterprise software at a mine site, and what internal resources are required to support it?
28. If we could build a LIMS that solved three specific problems for your lab tomorrow, what would they be?

Pro-Tip for the Meeting:

Do not just rapid-fire these questions. Use them as a guide. Let the client talk, and whenever they mention a pain point (e.g., "*Our scales keep disconnecting*" or "*The UI is too slow*"), dig deeper by asking: "*Can you give me a recent example of how that impacted a shift?*" This will yield the exact qualitative data you need to finalize your product features and business case.